Week 5 - Data Visualisation Critique

Hospitalised injuries from road crashes | Power BI dashboard | COS30045 | 104992981

1. Introduction

Purpose and audience

The Australian Government's Hospitalised injuries from road crashes Power BI dashboard presents 2011-2021 hospitalisation counts, crude rates per 100,000 population and a glossary (Department of Infrastructure, Transport, Regional Development, Communications and the Arts [DITRDCA], n.d.). It supports monitoring the burden of road-traffic injuries, relevant to the National Road Safety Strategy 2021-30 target of reducing serious injuries by at least 30% by 2030 (Australian Government, 2021).

Its primary users are transport and public-health analysts, researchers and regional road-safety officers who need defensible comparisons across years and population groups. For instance, an officer may ask whether higher remote-area rates reflect unequal injury burden or different population composition. Secondary users-journalists, advocates, students and the public-need readable scales and definitions before reporting findings. Typical questions concern high-risk road-user groups, trends, age and sex disparities, and differences between absolute counts and population rates.

2. Data Processing and Governance

Sources, definitions and transformations

The dashboard uses a custom extract from the Australian Institute of Health and Welfare National Hospital Morbidity Database, with Australian Bureau of Statistics population denominators (DITRDCA, 2023c). The statistical unit is an eligible hospital separation (an episode of care), not a unique crash or person. To reduce double counting, cases exclude transfers into another hospital and specified non-acute or statistical admissions; the road-traffic subset is identified using ICD-10-AM diagnostic and external-cause codes. When multiple external causes exist, coding generally uses the first relevant cause. The data dictionary also distinguishes cases ending in death from other cases; the headline display alone does not show how that exclusion is applied (DITRDCA, 2023c).

Rates are crude population rates rather than age-standardised risks or rates per distance travelled. Remoteness describes a patient's usual residence, not the crash location, and unknown demographic records may be omitted from subgroup outputs. Bed-day calculations follow different transfer rules from case counts, reducing direct comparability (DITRDCA, 2023c). These definitions materially change interpretation: the 2021 headline of 39,505 represents estimated hospitalisation cases, while the displayed remoteness bars sum to 38,876, leaving 629 cases outside the three plotted remoteness groups (DITRDCA, 2023b; Figure 1). The 2021 Remote rate of 224.6 per 100,000 must not be described as a measure of how dangerous remote roads are (Figure 2).

Quality, bias, privacy and data currency

Changes in reporting rules compromise straightforward comparisons over time. From July 2012, Victoria excluded cases managed entirely in emergency departments; a similar New South Wales policy change occurred in 2017. The data dictionary estimates that the Victorian change reduced the Australian count by approximately 2.8% in 2012. Six ICD-10-AM editions were also used across the series (DITRDCA, 2023c). Consequently, the reported increase across 2011-2021 must be interpreted cautiously: policy and coding changes may obscure part of the underlying trend, but the dashboard cannot establish its exact size.

The detailed data also create disclosure and fairness concerns. Small subgroup counts may risk disclosure if users combine filters; separate First Nations tables use suppressed entries, but a corresponding small-

number rule is not clearly explained on the displayed dashboard (DITRDCA, 2023a). First Nations injury data are available separately rather than as a main dashboard comparison, limiting visibility of an important equity dimension. The "intersex or indeterminate or missing" sex label combines unlike statuses, making missingness difficult to assess (Figure 4). In addition, the supplied dashboard ends in 2021, whereas the current data hub displays figures through 2023; a prominent data-period, revision and last-updated notice is needed to avoid confusing the historical dashboard with newer releases (National Road Safety Data Hub, 2026).

3. Visualisation Design

Data mapping and visual encoding

The dashboard combines line charts, categorical bars, a pie chart, pictograms and numeric cards. The following table evaluates their mappings against perceptual design guidance (Cleveland & McGill, 1984; Munzner, 2014).

View	Visual encoding	Critique
Trend	X = year; Y = count or rate	Appropriate for time, but the cropped range can visually exaggerate a modest percentage change; the scale must be explicit.
Age	Bars: count/rate by group	Bar baselines are clear. Unequal age intervals (40-64 spans 25 years; 17-25 spans nine, while 75+ is open-ended) weaken comparisons of raw counts.
Remoteness	Bar length and category	Direct labels help; the ranking changes from major-city counts to remote-area rates, so neither view tells the whole story.
Road user	Pie angle, area, colour	Angles are harder to compare accurately than common-position lengths; small slices lack legible labels.
Sex and KPI	Pictograms; large numerals	Identically sized icons do not encode the male-female rate difference. KPI cards support scanning but conceal subgroup spread.

The trend chart's non-zero baseline is not automatically invalid: a tight scale can help reveal variation in a line chart. Here, however, the 2011-2016 values rise from 34,033 to 38,963 (14.5%) while the plotted distance above the approximately 30,000 baseline increases disproportionately. Without prominent axis context, viewers could overestimate the trend (Figure 1, callout ①). An unexplained vertical separator near 2016 and a cryptic legend also make the known recording breaks hard to interpret. The pie's coloured angles are less effective than sorted bar lengths; road-user totals should include the small categories rather than depend on partly omitted slice labels (Cleveland & McGill, 1984).

Age-group counts peak in the wide 40-64 interval, whereas age-specific crude rates peak at 17-25 (Figures 1-2). A viewer could mistake the first for a higher individual risk. Similarly, major cities have the most admissions, but Remote residents have the highest per-person rate. These are differences between absolute burden and population-normalised rates, not contradictions. The central cards use quick-to-scan numbers, but an "average five bed days" card hides meaningful variation between road-user groups (Figure 1).

Colour, labels and layout

The navy-and-teal palette is visually cohesive, and most numeric cards are high contrast. Yet the same dark navy means total injuries in one chart, Major Cities in another and Driver in the pie, weakening semantic consistency. Several pie slices rely on hue rather than labels, which is problematic under the WCAG requirement that colour must not be the only way information is conveyed (World Wide Web Consortium [W3C], 2024). The dense, small Glossary text and abbreviations such as “POP’N” demand unnecessary interpretation. Key caveats are physically separated from the trend they qualify (Figure 4), contrary to at-a-glance dashboard communication (Few, 2013).

Interactivity and implementation

The visible slicers broadly follow Shneiderman’s (1996) overview–filter–details model, but have discoverability problems. Figure 3 shows an “Other or unknown” remoteness option that is not plotted as a separate bar, nested road-user options, a long counterparty menu with a partly clipped “Select all”, and an “Unknown” age choice even though unknown records are excluded from rates. The counts and rates tabs also provide different slicers without explaining why. A small Reset control and no obvious summary of active selections make combinations difficult to track.

Live dashboard exploration confirms useful chart linking (Figure 5A). Clicking an age-group bar or road-user pie slice changes or highlights related values in the left-hand time-series and remoteness charts. In the selected views, grey portions of bars preserve context while coloured portions emphasise the subgroup, although the difference between a filtered total and a highlighted subset is not explained. Hovering over the time-series line reveals the year and its value; one captured tooltip shows 4,526 for a selected subgroup in 2017 (Figure 5B). However, the tooltip can overlap a neighbouring KPI card. Unlike the age and road-user charts, the male/female pictograms in the rates tab are not directly clickable to filter the other views: users must use the Sex slicer. The rate charts also respond to selections (Figure 5C), but this inconsistency limits exploratory comparison. These observations verify linking and hover details, but not tab-to-tab selection persistence, interaction latency or small-screen responsiveness.

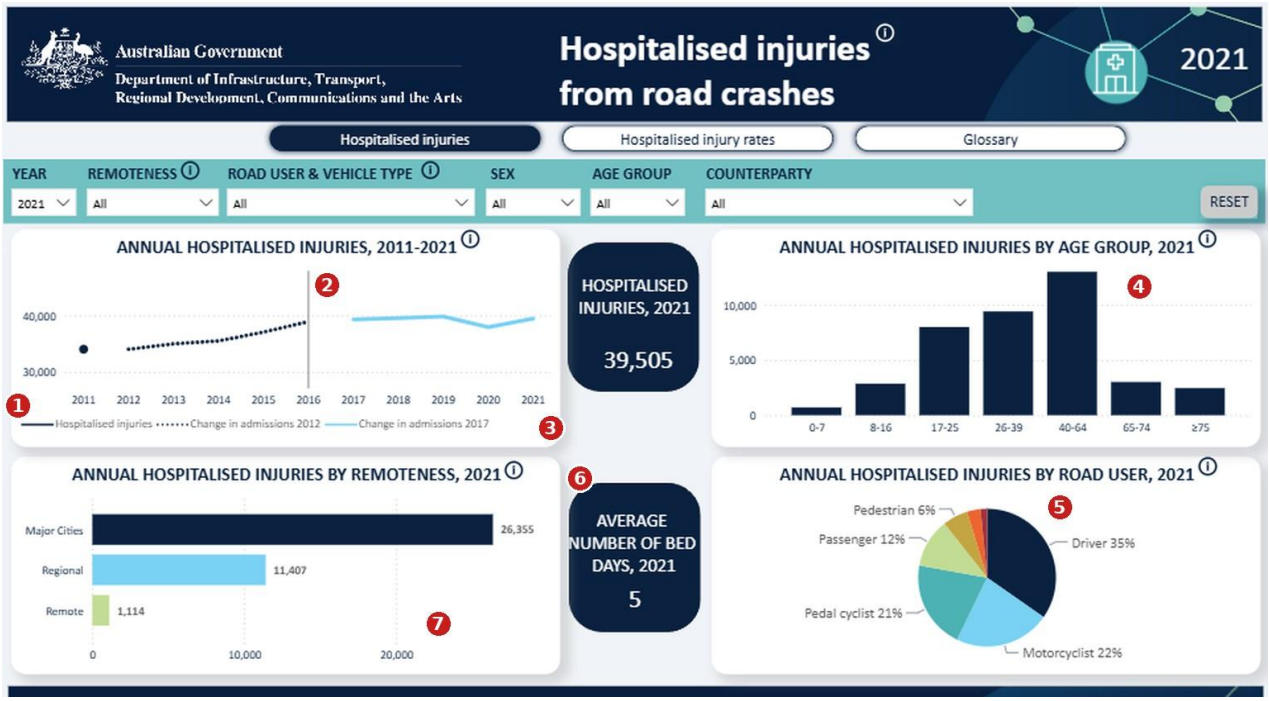


Figure 1. Counts view, annotated: ① tight y-axis; ② unexplained vertical divider; ③ ambiguous legend; ④ wide age bins; ⑤ small unlabeled pie slices; ⑥ mean bed-days KPI; ⑦ remoteness totals do not include all cases.



Figure 2. Rates view, annotated: ① restricted y-axis; ② younger peak rate; ③ Remote rate exceeds Major Cities; ④ equal-size sex pictograms; ⑤ cramped KPI label; ⑥ filters differ from counts view.

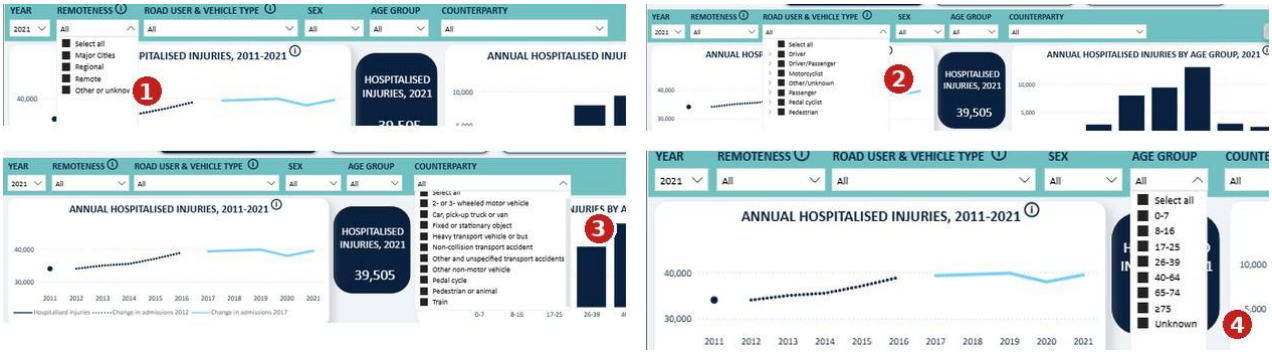


Figure 3. Filter dropdown screenshots: ① selectable unknown remoteness; ② nested road-user options; ③ long counterparty list with clipped select-all control; ④ unknown age choice.

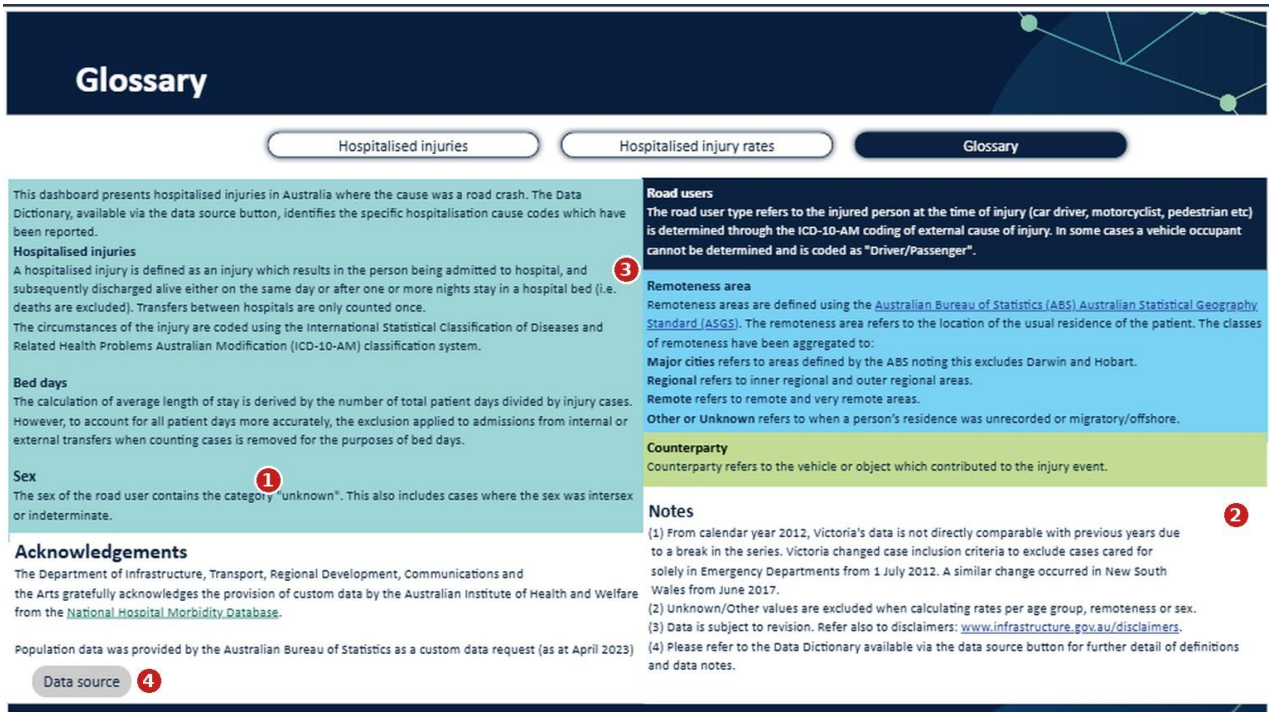


Figure 4. Glossary screenshot: ① conflated sex/unknown description; ② important series-break notes placed low on the page; ③ dense explanatory panels; ④ source information without a visible update date.



Figure 5. Evidence from live dashboard exploration: (A) selecting age groups changes or highlights linked visuals; (B) a 2017 line-chart hover shows the selected subgroup value; (C) selection affects rate charts, whereas the sex pictograms cannot be clicked to initiate the same interaction.

4. Recommendations

1. **Clarify the time series.** Keep a detailed line scale only if it is clearly labelled, add an accessible zero-based overview or percentage-change alternative, and mark the Victoria 2012 and NSW 2017 recording breaks directly on the chart. Explain classification changes beside the graph rather than only in the Glossary.
2. **Replace weak encodings.** Use sorted, labelled road-user bars in place of the pie, retaining small or unknown groups. For comparisons across years, use aligned road-user trend lines. Replace identical male/female pictograms with keyboard-accessible, clickable labelled bars that encode both rates and cross-highlight other charts, matching the behaviour of the interactive age bars.
3. **Compare like with like.** Place subgroup counts beside crude rates and explain that remoteness is usual residence, not crash location. Where data support it, provide standardised rates and comparable age categories; otherwise highlight unequal age-bin widths and the open-ended oldest group.
4. **Make data limitations explicit.** Reconcile the missing-remoteness cases with a labelled "Other/unknown" line, footnote or bar. Surface missingness, the definition of cases versus bed days, the small-cell suppression policy, coverage period and revision date. Link to the updated 2023 reporting, clearly distinguished from this 2011-2021 view.
5. **Improve filter discoverability.** Preserve the useful linked chart highlighting, but visibly distinguish selected subgroups from overall totals and add a clear-selection action. Display active slicer choices, make "Reset all" more prominent, provide search in long lists and prevent clipped controls. Label subgroup context in tooltips and explain why the counts and rates tabs have different filter choices.

6. **Test accessibility and implementation.** Ensure direct labels, keyboard-accessible chart interactions and WCAG 2.2 contrast; provide a linked First Nations view with appropriate disclosure safeguards. Test chart responsiveness, tooltip placement, tab persistence and narrow-screen layouts with specialist and general users. For example, time how easily each group can compare a selected road user with all users, recover the original view, and explain what a grey-highlighted bar means.

Conclusion

The dashboard offers valuable chart linking and on-demand details, but inconsistent interaction, ambiguous highlights and hard-to-find data caveats weaken its usefulness. Clarifying definitions and selection states, improving encodings and extending direct interaction to subgroup comparisons would support more reliable interpretation by both expert and non-expert audiences.

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